Two Acts

🤘 Cache Me If You Can

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2. <some time passes >

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What's happening in the model?

Both Prefill and Decode perform forward passes through the model.
The two phases have different characteristics.

Words we will hear all morning

Token: a unit of information used by an LLM, usually a word or part of a word.

- Tokens can also represent image content or serve as structural markers (e.g. end of sequence or thinking markers).

Time to first token (TTFT): time between sending the prompt and receiving the first token.

Tokens per second per user: how quickly new tokens are generated.

Phase One: Generating internal state

During prefill, the model reads **the whole prompt** chunk by chunk and builds its internal state.

In every layer, a matrix of prompt tokens is multiplied by layer weights.
Weights are read from memory **once** and used for every token of the prompt.

The duration of this prefill phase determines **Time To First Token**.

Phase Two: Decode

During decode, the answer is written **one token at a time**.
For each token, a **full pass** through the model is performed.

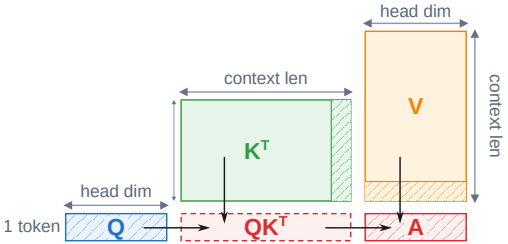
- On every pass, all weights are reread from memory.

How long decode takes determines **tokens per second**.

Refresher: How the attention mechanism works in decode

The standard attention formula is

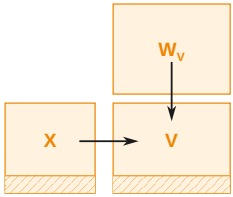
$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$



Q, K, V are created with linear projections.

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

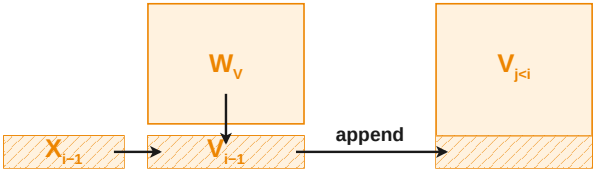
The cost of projections increases with context length.



Introducing KV-cache

In text transformers, past data in K and V does not depend on future tokens.

- **Idea**: We can store calculated K and V for subsequent iterations.
- We call this a KV cache.



KV caching avoids repeated calculations but increases memory usage.

- With long contexts (e.g. 1M tokens), the KV cache can be massive.
- Many models (e.g. DeepSeek) try to minimize KV-cache size per token.

Note that cost of attention still increases with context length.

Compute-bound and memory-bound

Before cores can use weights, the weights must travel from memory (DRAM) to the cores.

Prefill is applied to many tokens in parallel.

- Weights can be loaded once for hundreds of tokens.
- With fewer DRAM accesses, the limit is the speed of calculation.
- We call this **compute-bound**.

Decode can only generate 1 token at a time.

- Weights are reloaded from DRAM for each pass.
- With many DRAM accesses, the limit is how fast we can read from memory.
- We call this **memory-bound**.

Despite using the same model and hardware, the two phases have different limits.

Calculating the speed limit

With one user, each decode step reads all weights once.

$$\textbf{minimum token time} = \frac{\text{bytes read per step}}{\text{memory bandwidth}}$$

Example: generating one token reads 1.8 GB of weights at 288 GB/s (one N150):

- 6.2 ms per token $\approx$ 160 tokens/second.

To go faster, the model must either:

- Reduce data size with smaller number formats.
- Load data less often (e.g. batching or speculative decoding).

Objectives of Act I

The goals of Act I are:

1. Run a small DeepSeek model and generate tokens.
2. Measure time spent on each phase
   - At various sequence lengths.
3. Determine the speed limits of models on your card and compare them with measurements.